Strategy Design Report

Multi-Factor Dollar-Neutral European Equity Strategy

QFAT Group Project — Investment Committee Presentation

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1 Executive Summary

We design a **quantitative equity long-short strategy** that combines multiple alpha signals into a single composite score. The strategy goes long the top quintile and short the bottom quintile of European equities (Netherlands, France, Belgium), rebalanced monthly, with inverse-volatility weighting.

Key results (net of 30 bps one-way transaction costs):

Metric	Value
Annualised CAPM Alpha	+8.73%
Information Ratio (CAPM)	0.747
Sharpe Ratio	0.732
Max Drawdown	−35.54%
Win Rate	64.6%
CAPM Beta	−0.025

The strategy selects **four signals** (Value, Quality, Conservative Investment, and Momentum) through a 5-fold chronological cross-validation procedure designed to mitigate model-selection overfitting, dropping the Low Risk (BAB) signal which did not improve out-of-fold IR. An in-sample CAPM IR of 0.747 net of costs is economically strong in-sample, though likely optimistic relative to live implementation.

Why this strategy?

- **Assignment constraint:** Pure momentum strategies are not permitted. Our momentum signal enters with only ∼25% weight in a multi-factor composite—this is a *combination strategy*, not a momentum strategy.
- **Evaluation criterion:** The Information Ratio vs. the CAPM (MktRF) is the primary performance metric. Our strategy is dollar-neutral (equal capital on each side) with near-zero realised market beta, so the IR measures the risk-adjusted alpha after controlling for market exposure.
- **No look-ahead bias:** All signals use only information available at the time of portfolio formation.

2 Data & Universe

2.1 Data Sources

We work with four datasets:

Dataset	Rows	Firms	Period	Key Variables
Monthly stock data	∼254,000	1,905	Jun 1991–May 2022	RET, RET11, ME, b, h, s, ivol
Annual fundamentals	∼23,000	1,905	1990–2020	BEME, OP, INV
Firm characteristics	1,905	—	—	ISIN, country
European FF Factors	373 mo.	—	Jun 1991–Jun 2022	MktRF, SMB, HML, WML, RF

2.2 Universe Composition

Country	# Firms	% of Universe
France	1,440	75.6%
Netherlands	250	13.1%
Belgium	215	11.3%

The universe is dominated by French equities, with the Netherlands and Belgium providing additional diversification. The cross-country composition provides exposure to the broader Eurozone economy while maintaining data quality within a coherent regulatory environment.

2.3 Variable Definitions

Variable	Definition
RET	Total stock return in month t
RET11	Past 11-month return supplied in the dataset, capturing medium-term momentum while avoiding s
ME	Market equity (price × shares outstanding)
b	Market beta supplied in the dataset, estimated from daily returns
ivol	Idiosyncratic volatility (residual std. dev.)
BEME	Book-to-market equity ratio
OP	Operating profitability
INV	Asset growth rate ($\mathrm{TA}_t/\mathrm{TA}_{t-1} - 1$)

2.4 Fama–French Lag Convention

To prevent look-ahead bias when using annual fundamentals (BEME, OP, INV), we follow the standard **Fama–French lag convention**:

$$\mathrm{fyear}_{\mathrm{use}} = \begin{cases} \mathrm{year} - 1 & \text{if month} \geq 7, \\ \mathrm{year} - 2 & \text{if month} < 7. \end{cases}$$

This assumes fiscal year t financial statements are publicly available by July of year $t+1$, which is conservative and standard in the literature (Fama & French, 1993).

2.5 Forward Return Construction

Signals are observed at the end of month t. The portfolio is formed at the end of month t and held during month $t+1$. The forward return is:

$$R_{i,t}^{\text{fwd}} = R_{i,t+1}$$

We include a **gap guard**: if a firm is missing from the data in month $t+1$ (i.e., the next observation is not the immediately following calendar month), we set $R_{i,t}^{\text{fwd}} = \text{NaN}$ to avoid stale return leakage.

3 Signal Design & Economic Rationale

We begin with a universe of **five candidate signals**, each backed by a well-documented economic mechanism. The signals are designed to capture distinct dimensions of the cross-section of expected returns.

3.1 Value (BEME)

Direction: High BEME = cheap = good (long high, short low).

The value premium is one of the most studied anomalies in finance. High book-to-market firms are either:

- **Distressed** firms that carry higher risk, compensating patient investors (risk-based explanation; Fama & French, 1993), or
- **Over-sold** firms where the market over-extrapolates recent bad news, creating mean-reversion opportunities (behavioural explanation; De Bondt & Thaler, 1985; Lakonishok, Shleifer & Vishny, 1994).

Who pays us? Growth-chasing investors who bid up glamour stocks. Benchmark-constrained funds forced to sell distressed names near index rebalancing dates.

Initial weight: 25%—the most robust and well-documented anomaly.

3.2 Quality (OP)

Direction: High OP = profitable = good (long high, short low).

Firms with high operating profitability earn higher expected returns because:

- High profitability signals **economic moat** and competitive advantage, which the market underprices (Novy-Marx, 2013).
- Unprofitable firms attract **lottery-seeking** retail investors who overpay for upside optionality (Bali, Cakici & Whitelaw, 2011).

Who pays us? Lottery-seeking retail investors. Investors who chase revenue growth without regard to profitability.

Initial weight: 20%

3.3 Conservative Investment (−INV)

Direction: Low INV = conservative = good (long low-investment, short high-investment).

Firms that invest aggressively tend to earn lower subsequent returns because:

- **Empire-building managers** over-invest, destroying shareholder value (Titman, Wei & Xie, 2004).
- Markets **over-extrapolate** past growth, leading to overvaluation of high-growth firms (Cooper, Gulen & Schill, 2008).

Who pays us? Investors who extrapolate past asset growth into the future. Growth-oriented funds that systematically overpay for expanding firms.

Initial weight: 15%—weakest standalone signal, but provides valuable diversification.

3.4 Low Risk / BAB ($-\beta$)

Direction: Low beta = safe = good (long low-β, short high-β).

The Betting Against Beta (BAB) factor exploits the finding that the Security Market Line is **too flat**—low-beta stocks earn higher risk-adjusted returns than high-beta stocks (Frazzini & Pedersen, 2014). This occurs because:

- **Leverage-constrained investors** (mutual funds, pension funds) cannot lever up low-beta stocks, so they tilt toward high-beta stocks. This excess demand pushes high-beta prices up and future returns down.
- The effect is persistent because it is rooted in **institutional constraints** (leverage limits, benchmarking mandates) rather than behavioural biases.

Who pays us? Leverage-averse pension funds. Benchmarked funds that need to match index beta. Retail investors who prefer high-beta "exciting" stocks.

Initial weight: 20%

3.5 Momentum (RET11)

Direction: High past return = good (long winners, short losers).

Stocks that have performed well over the past 2–12 months tend to continue performing well (Jegadeesh & Titman, 1993). The mechanism is:

- **Under-reaction** to firm-specific news: information diffuses slowly (Hong & Stein, 1999).
- **Disposition effect:** investors sell winners too early and hold losers too long (Grinblatt & Han, 2005).
- **Slow-moving capital:** arbitrageurs cannot immediately correct mispricings (Mitchell, Pedersen & Pulvino, 2007).

Important note: The RET11 variable skips the most recent month ($t-1$) to avoid the well-documented **short-term reversal** effect (Jegadeesh, 1990).

Initial weight: 20%—capped to ensure this is not a pure momentum strategy (assignment constraint).

3.6 Why These Five Signals?

The five signals span the major dimensions of the cross-section of expected returns:

Critically, these signals have **low pairwise correlation** (especially value vs. momentum, which are negatively correlated), providing substantial diversification when combined. This is the core advantage of a multi-factor approach.

Dimension	Signal	Key Paper(s)
Value	BEME	Fama & French (1993, 2015)
Profitability	OP	Novy-Marx (2013), Fama & French (2015)
Investment	INV	Cooper, Gulen & Schill (2008), Fama & French (2015)
Low Risk	β	Frazzini & Pedersen (2014)
Momentum	RET11	Jegadeesh & Titman (1993), Asness et al. (2013)

4 Signal Construction Methodology

4.1 Cross-Sectional Z-Scores

For each signal k and each month t, we compute a cross-sectional z-score:

$$z_{i,t}^{(k)} = \frac{X_{i,t}^{(k)} - \bar{X}_t^{(k)}}{\sigma_t^{(k)}}$$

where $X_{i,t}^{(k)}$ is the raw signal value for stock i in month t, and $\bar{X}_t^{(k)}$, $\sigma_t^{(k)}$ are the cross-sectional mean and standard deviation.

Why z-scores?

- They make signals **comparable** across different scales (BEME is a ratio, OP is a percentage, RET11 is a cumulative return).
- They are **time-invariant**: each month's distribution is standardised, preventing any single signal from dominating due to wider dispersion.

4.2 Winsorization

Before computing z-scores, we **winsorize** each signal at the 1st and 99th percentiles:

$$X_{i,t}^{(k),w} = \text{clip}\Big(X_{i,t}^{(k)},\ Q_{0.01},\ Q_{0.99}\Big)$$

This limits the influence of extreme outliers without discarding them entirely. Winsorization is standard practice in empirical asset pricing.

4.3 Signal Direction

For signals where **lower is better** (INV, β), we negate the z-score:

$$z_{i,t}^{\text{invest}} = -z(\text{INV}_{i,t}), \qquad z_{i,t}^{\text{lowrisk}} = -z(\beta_{i,t})$$

This ensures that for all signals, **higher z-score = more attractive stock**.

4.4 Composite Score

The composite score is a weighted sum of individual z-scores:

$$Z_{i,t} = \sum_k w_k \cdot z_{i,t}^{(k)}$$

Signal	Weight	Rationale	Key Paper(s)
Value	25%	Most extensively documented anomaly	Fama & French (1993, 2015); LSV (19
Quality	20%	Strong standalone; complements value	Novy-Marx (2013); Fama & French (2
Low Risk	20%	Structural institutional constraints	Frazzini & Pedersen (2014)
Momentum	20%	Well-documented; capped at 20%	Jegadeesh & Titman (1993); AMP (20
Investment	15%	Weakest standalone; diversification	Cooper et al. (2008); Fama & French (

where the weights w_k are set as initial proportions and later refined through model selection (Section 5).

Initial (full-model) weights:

How were these weights determined? The weights are **not sourced from any single paper**—no academic paper prescribes exact signal weights for a multi-factor composite. Instead, they reflect an **economic judgment** based on each factor's documented robustness:

- Most multi-factor papers (e.g., the Fama–French 5-factor model) use **equal-weighted** factors. Our slight tilt toward value (25% vs. 20%) reflects its status as the single most replicated anomaly across geographies and decades.
- Momentum is **deliberately capped** at 20% to prevent classification as a pure momentum strategy under the assignment rules.
- Investment receives the lowest weight (15%) because its standalone long-short IR is the weakest in European data, though it still adds diversification value.
- Quality and Low Risk receive equal 20% weights, reflecting comparable standalone evidence.

Importantly, these initial weights are only a **starting point**. The model selection procedure (Section 5) determines *which* signals to retain, and the rescaling function proportionally adjusts the survivors so they sum to 1. The final weights used in the backtest are a mechanical consequence of the selection outcome, not hand-tuned to the data.

4.5 Coverage

After merging monthly and annual data, computing z-scores, and requiring non-missing forward returns and positive idiosyncratic volatility, the investable universe contains approximately:

- ∼130,000 valid firm-months
- ∼450 stocks per month on average
- Coverage spans June 1992 to April 2022

5 Model Selection Procedure

5.1 Motivation

While all five signals are theoretically motivated, not all may add value *in this specific dataset*. Including a signal that adds noise without improving risk-adjusted returns dilutes the composite score. We therefore conduct a **data-driven model selection** to determine which subset of signals maximises the Information Ratio.

5.2 Design Principles

1. **Fix base signals:** Value and Quality are always included. These are the two most fundamental anomalies. Removing them would depart too far from established theory.
2. **Optional signals:** Investment, Low Risk, and Momentum are candidates to be included or excluded.
3. **Combinatorial search:** We evaluate all $2^3 = 8$ possible subsets of the optional signals, always combined with Value and Quality.
4. **Proportional rescaling:** When a signal is excluded, remaining weights are rescaled proportionally:

$$w_k^{\text{rescaled}} = \frac{w_k^{\text{original}}}{\sum_{j \in \text{selected}} w_j^{\text{original}}}$$

5.3 K-Fold Time-Series Cross-Validation

To mitigate overfitting to the full in-sample period, we use a **k-fold chronological split**:

1. Divide the full backtest period into $k = 5$ equal, contiguous, non-overlapping time folds.
2. For each signal combination, run the full backtest over the entire period.
3. Compute the CAPM Information Ratio **separately within each fold**:

$$\text{IR}_{\text{fold}} = \frac{\hat{\alpha}_{\text{fold}}}{\hat{\sigma}(\epsilon)_{\text{fold}}} \times \sqrt{12}$$

4. The candidate's score is the **average IR across all k folds**.
5. Select the signal combination with the **highest average fold IR**.

Why k-fold rather than a single train/test split?

- A single split is sensitive to the choice of cutoff date. K-fold averages over multiple sub-periods, giving a more robust estimate.
- We use **chronological folds** to respect the time-series nature of financial data—no future information leaks into past folds.

5.4 Candidate Evaluation Results

The 8 candidate combinations and their 5-fold average IRs:

#	Signals	Avg IR (5-fold)
1	Value, Quality	0.374
2	Value, Quality, Investment	0.413
3	Value, Quality, Low Risk	−0.027
4	Value, Quality, Momentum	0.850
5	Value, Quality, Investment, Low Risk	0.082
6	**Value, Quality, Investment, Momentum**	**0.866**
7	Value, Quality, Low Risk, Momentum	0.491
8	Value, Quality, Investment, Low Risk, Momentum	0.457

Combination 6 (Value, Quality, Investment, Momentum) achieves the highest average fold IR at 0.866, with positive IR in all five folds. Adding Low Risk consistently lowers the average (combinations 3, 5, 7, 8 vs. their Low-Risk-free counterparts), confirming that the BAB signal adds noise in this universe.

5.5 Result: Selected Model

The model selection procedure selects **Value, Quality, Investment, and Momentum**—dropping the Low Risk (BAB) signal.

Final rescaled weights:

Signal	Original Weight	Rescaled Weight
Value	0.25	**0.3125** (31.25%)
Quality	0.20	**0.2500** (25.00%)
Investment	0.15	**0.1875** (18.75%)
Momentum	0.20	**0.2500** (25.00%)
Total	0.80	**1.0000**

5.6 Why Was Low Risk Dropped?

Several possible explanations:

- **BAB in Europe:** The BAB premium has been documented primarily in U.S. equities. In the NL/FR/BE universe, the cross-sectional spread in beta may be narrower.
- **Correlation with other signals:** Low-beta stocks tend to overlap with high-quality, low-investment firms. Once Quality and Investment are included, Low Risk's marginal contribution may be small.
- **Estimation noise:** Monthly beta estimates from daily returns within a single month are inherently noisy, diluting signal effectiveness.
- **Fold-validation guards against overfitting:** Even though the full-sample BAB spread may look reasonable, it does not consistently add fold-by-fold IR.

6 Portfolio Construction

6.1 Quintile Sort (Long-Short)

Each month, stocks with valid composite scores are sorted into **quintiles** based on Z:

- **Long leg:** Top quintile (Q5)—the ∼20% of stocks with the highest composite score.
- **Short leg:** Bottom quintile (Q1)—the ∼20% with the lowest composite score.

With ∼450 valid stocks per month, each leg contains approximately **90 stocks**, providing substantial diversification.

Minimum stock requirement: At least 20 stocks per leg. If either leg has fewer, we skip that month.

6.2 Inverse-Volatility Weighting

Within each leg, stocks are weighted by the **inverse of their idiosyncratic volatility**:

$$w_i = \frac{1/\sigma_{\epsilon,i}}{\sum_{j \in \text{leg}} 1/\sigma_{\epsilon,j}}$$

where $\sigma_{\epsilon,i}$ is the idiosyncratic volatility of stock i.

Why inverse-volatility weights?

- **Reduces noise:** Stocks with high idiosyncratic volatility have less predictable returns. Down-weighting them reduces portfolio variance.
- **Improves Sharpe ratio:** Inverse-volatility weighting is a pragmatic risk-balancing heuristic; it is not the exact minimum-variance solution, but it tilts the portfolio toward lower-risk names, improving risk-adjusted returns in practice.
- **Avoids concentration:** Unlike equal weighting, inverse-vol weighting naturally limits exposure to small, volatile stocks.

Note: Signal weights (from k-fold selection) and inverse-vol weights operate at different stages. Signal weights determine *which stocks are attractive* (the composite score used for ranking). Inverse-vol weights determine *how much of each stock to hold* within the selected quintile. The former is about signal combination; the latter is about position sizing.

6.3 Dollar Neutrality

The strategy is **dollar neutral**: the long leg sums to 1 and the short leg sums to 1, for a total gross exposure of 2 per 1 of AUM. The portfolio return is:

$$R_{\text{L/S},t} = R_{\text{long},t} - R_{\text{short},t}$$

6.4 Transaction Costs

We assume **one-way transaction costs of 30 basis points** as a plausible implementation benchmark for European mid-cap equities:

- Half the bid-ask spread (∼10–20 bps for mid-cap European stocks)
- Brokerage commissions (∼5 bps)
- Market impact (∼5–10 bps for moderate position sizes)

Turnover calculation: Each month, we compute the total absolute change in weights:

$$\text{Turnover}_t = \sum_{i \in \text{long}} |w_{i,t} - w_{i,t-1}| + \sum_{j \in \text{short}} |w_{j,t} - w_{j,t-1}|$$

The transaction cost deduction is:

$$\text{TC}_t = \text{Turnover}_t \times 0.0030$$

And the net return is:

$$R_{\text{L/S, net},t} = R_{\text{L/S},t} - \text{TC}_t$$

7 Backtest Results

7.1 Performance Summary

All returns below are computed on a *self-financing* 100/100 long-short book (200% gross exposure), reported per unit of equity capital. This is the standard academic convention for long-short factor portfolios.

Metric	Gross	Net (30 bps)
Ann. Return	+13.13%	+8.56%
Ann. Volatility	11.67%	11.69%
Sharpe Ratio	1.125	0.732
Sortino Ratio	1.299	0.879
Max Drawdown	−31.65%	−35.54%
Win Rate	71.6%	64.6%
CAPM Alpha (ann.)	+13.30%	+8.73%
CAPM Beta	−0.025	−0.025
Information Ratio	**1.140**	**0.747**

7.2 CAPM Regression

The primary evaluation metric is the Information Ratio from the CAPM regression:

$$R_{\mathrm{L/S},t} = \alpha + \beta \cdot \mathrm{MktRF}_t + \epsilon_t$$

- **Alpha:** The intercept α represents the risk-adjusted return not explained by market exposure. Our monthly alpha of 0.73% (net) is statistically significant with a Newey–West t-statistic of 3.38 (using HAC standard errors with 6 lags to correct for autocorrelation).
- **Beta ≈ -0.025:** Near-zero market exposure, confirming that the dollar-neutral construction achieves negligible directional risk in practice.
- **Information Ratio:**
$$\mathrm{IR} = \frac{\alpha}{\sigma(\epsilon)} \times \sqrt{12} = 0.747$$
An IR above 0.5 is generally considered strong.

7.3 4-Factor Decomposition (MktRF, SMB, HML, WML)

Consistent with the project brief, we evaluate the strategy primarily by its CAPM Information Ratio. We also report a 4-factor alpha as a robustness and attribution check, since part of the CAPM alpha may reflect exposure to standard value and momentum premia.

To decompose the CAPM alpha into known factor exposures versus residual return, we regress on a 4-factor model using MktRF, SMB, HML, and WML:

$$R_{\mathrm{L/S},t} = \alpha_{\mathrm{FF4}} + \beta_1 \,\mathrm{MktRF}_t + \beta_2 \,\mathrm{SMB}_t + \beta_3 \,\mathrm{HML}_t + \beta_4 \,\mathrm{WML}_t + \epsilon_t$$

Key takeaways:

Factor	Coefficient	Std. Error	t-stat	p-value
α (monthly)	+0.0049	0.002	2.76	0.006
MktRF	−0.029	0.037	−0.80	0.425
SMB	+0.089	0.078	1.14	0.256
HML	+0.417	0.069	6.07	0.000
WML	+0.140	0.048	2.93	0.004

Table 1: 4-factor regression on MktRF, SMB, HML, and WML (net of TC).

- **FF4 alpha** annualises to +5.89% ($t = 2.76$, $p = 0.006$), which is both economically meaningful and statistically significant at the 1% level. The FF4 Information Ratio is 0.530.
- **HML loading** of +0.42 ($t = 6.07$) confirms the strategy's strong value tilt—expected given the 31.25% value weight.
- **WML loading** of +0.14 ($t = 2.93$) reflects the 25% momentum component.
- **MktRF and SMB** loadings are statistically indistinguishable from zero, confirming negligible market and size exposure.
- Comparing CAPM alpha (+8.73%) to FF4 alpha (+5.89%), part of the CAPM alpha is attributable to standard value and momentum factor exposures. This is expected, since the strategy is explicitly constructed using value- and momentum-related signals. However, a substantial positive alpha remains after controlling for MktRF, SMB, HML, and WML, suggesting incremental return beyond simple benchmark factor replication.

7.4 Interpretation of Results

- **Near-zero market beta:** CAPM beta of ≈ -0.025 means virtually no directional market exposure. Note that this is an *outcome* of diversified multi-factor construction, not an explicit beta-hedging constraint—the portfolio is dollar-neutral, not beta-neutral by design.
- **Alpha is statistically significant:** Newey–West t-statistic of 3.38 for net alpha.
- **Transaction costs are material:** IR drops from 1.14 (gross) to 0.75 (net), a ∼34% reduction.
- **Positive win rate:** 64.6% winning months substantially outnumber losing months.

8 Robustness & Sensitivity Analysis

8.1 Sub-Period Stability

A strategy that works only in one sub-period is likely overfit. We split the backtest into halves and decades:

Period	n (months)	Ann. Return	Sharpe	IR
Full sample	370	+8.56%	0.732	0.747
First half (1992–2006)	185	+11.25%	0.849	0.884
Second half (2007–2022)	185	+5.87%	0.595	0.602
1990s	114	+8.91%	1.185	1.296
2000s	120	+11.66%	0.717	0.741
2010s	120	+3.59%	0.382	0.325

IR is positive in every sub-period. The first half is stronger than the second, and the 2010s are the weakest decade—consistent with the well-documented compression of factor premia in recent years. Nonetheless, the strategy remains profitable throughout.

8.2 Crisis Performance

A dollar-neutral long-short strategy may be more resilient than a long-only portfolio during market crises, but crisis performance depends on factor exposures and crowding:

Crisis	Strategy (cum.)	Market (cum.)
Dot-com crash (2000–02)	+147.6%	−44.8%
GFC (2007–09)	+6.2%	−29.2%
Euro crisis (2011–12)	−4.4%	+4.9%
COVID crash (2020 H1)	−25.3%	−12.2%

The strategy performed well during the dot-com crash and GFC, but lost money during the COVID sell-off—likely due to a sharp momentum reversal. The Euro-crisis loss is modest. Low CAPM beta helps insulate from broad market drawdowns, but factor-specific risks (especially momentum crashes) can still cause material losses.

8.3 Signal Weight Sensitivity

We test six alternative weighting schemes to verify the strategy is not fragile to the specific weight choices:

All six variants produce positive IR, ranging from 0.43 (no momentum) to 0.75 (baseline). This confirms that the alpha comes from the underlying signals, not from a precisely tuned weight vector. Removing momentum entirely roughly halves the IR, indicating that momentum is a meaningful contributor but not the sole driver.

8.4 Individual Signal Decomposition

We run each signal as a standalone long-short strategy and compare to the composite.

Variant	Weights (Val / Qual / Inv / Mom)	IR
Baseline	31% / 25% / 19% / 25%	0.747
Equal-weight	25% / 25% / 25% / 25%	0.721
Value-tilt	40% / 20% / 15% / 25%	0.713
Quality-tilt	20% / 40% / 15% / 25%	0.658
Momentum-tilt	20% / 20% / 15% / 45%	0.650
No momentum	40% / 35% / 25% / 0%	0.429

Key insight: The composite achieves a **higher IR than any individual signal**, demonstrating the **diversification benefit** of combining weakly correlated alpha sources. Each included signal contributes positively—none is pure noise.

9 Risk Analysis & Failure Modes

9.1 Key Risks

Risk	Description	Mitigation
Factor crowding	Many quant funds trade similar signals → premia compress	Multi-signal diversification; smaller European universe less crowded
Momentum crash	Sudden reversals (e.g., Mar 2009)	Momentum is only 25%; value provides natural hedge
Value trap	Cheap stocks continue to deteriorate	Quality signal filters out unprofitable cheap stocks
Short squeeze	Concentrated shorts in stressed markets	∼90 stocks per short leg; inverse-vol limits concentration
Regime change	Structural shifts in factor premia	Four signals across different dimensions; k-fold tests persistence
Transaction costs	Higher-than-modelled TC erodes alpha	Positive IR even at 30 bps; monthly rebalancing realistic
Data quality	Missing data, survivorship bias	Conservative NaN handling; gap guards on forward returns

9.2 Max Drawdown Analysis

The worst drawdown (net) is approximately -35.5%. This is substantial for a dollar-neutral long-short strategy, but it is economically plausible given the portfolio structure. The strategy is a self-financing 100/100 long-short book with 200% gross exposure, so factor-level losses are mechanically amplified relative to a long-only portfolio. Moreover, near-zero market beta does not eliminate drawdown risk: the portfolio remains exposed to value drawdowns, momentum reversals, crowding, short squeezes, and regime shifts. In our case, the 4-factor regression indicates meaningful HML and WML exposure, so periods of severe underperformance in value and momentum can generate deep drawdowns even when directional market exposure is minimal. The drawdown should therefore be interpreted not as evidence that the strategy is invalid, but as the cost of harvesting these premia in a levered long-short format. It also suggests that any live implementation would require disciplined sizing, realistic leverage limits, and potentially a volatility-targeting or drawdown-control overlay.

9.3 Capacity Constraints

- Universe of ∼450 liquid stocks across 3 countries
- Average ∼90 stocks per leg with inverse-vol weighting
- Capacity is likely limited by the relatively narrow three-country universe and monthly turnover; a precise estimate would require ADV-based trading-capacity analysis

10 Conclusion

10.1 Summary of Design Choices

Design Decision	Choice	Justification
Signal universe	5 candidates	Cover major asset pricing dimensions
Model selection	K-fold CV (k=5), fix Value+Quality	Mitigate overfitting; anchor to theory
Selected model	Value, Quality, Inv., Momentum	Highest avg fold IR; Low Risk dropped
Weighting scheme	Proportional rescaling	Preserves relative economic judgment
Sort method	Quintile sort (top/bottom 20%)	Balance signal strength vs. diversification
Position sizing	Inverse idiosyncratic volatility	Reduce noise, improve risk-adj. returns
Transaction costs	30 bps one-way	Plausible benchmark for European mid-cap
Rebalancing	Monthly	Capture momentum; manageable turnover

10.2 Why This Strategy Works

1. **Diversification across alpha sources:** Four signals with low mutual correlation produce a composite more stable than any individual signal.
2. **Economic grounding:** Each signal is backed by decades of academic research and clear economic mechanisms—this is not data mining.
3. **Disciplined model selection:** The k-fold procedure helps reduce in-sample overfitting, though some post-selection bias remains.
4. **Partially implementation-aware:** Dollar-neutral construction, conservative TC, and sufficient diversification (∼90 stocks per leg)—though borrow fees, margin financing, and short-sale frictions are omitted.

10.3 Limitations & Caveats

- **Backtests are not live returns.** Transaction costs, short-selling constraints, and market impact may differ in practice.
- **Transaction cost model is partial.** The 30 bps one-way cost covers proportional turnover costs (spread, commission, and estimated market impact) but does *not* include stock-borrow fees on the short leg, margin financing costs, or tax frictions. In practice, these omitted costs could add an estimated 75–125 bps per year of additional drag, particularly for hard-to-borrow names. Additionally, the 100/100 long-short book requires 200% gross exposure; a prime broker would require initial margin on both legs, so the realised return on committed capital would be lower than reported.
- **Mild model selection bias.** The reported in-sample IR is measured *after* selecting the best-performing subset from 8 signal combinations via k-fold cross-validation. Although the search space is deliberately small and the cross-validation procedure mitigates overfitting, some upward bias in the selected model's in-sample IR is unavoidable.
- **Survivorship bias cannot be fully ruled out.** The dataset contains 1,905 unique firms versus ∼700 active in any given month, suggesting that delisted firms are at

least partially included. However, we cannot verify whether all delistings—and their terminal returns—are fully captured. The long-short construction partially mitigates this concern, since survivorship bias inflates both legs.

- **The sample period (1991–2022)** includes multiple regimes but may not represent all future conditions.
- **Factor premia may decay** as more capital flows into quantitative strategies.
- **Frictionless short-selling is assumed**—in practice, short-sale costs and availability may vary.

11 References

- Ang, A., Hodrick, R.J., Xing, Y. & Zhang, X. (2006). The cross-section of volatility and expected returns. *Journal of Finance*, 61(1), 259–299.
- Asness, C.S., Moskowitz, T.J. & Pedersen, L.H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3), 929–985.
- Bali, T.G., Cakici, N. & Whitelaw, R.F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2), 427–446.
- Cooper, M.J., Gulen, H. & Schill, M.J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4), 1609–1651.
- Daniel, K., Hirshleifer, D. & Sun, L. (2020). Short- and long-horizon behavioral factors. *Review of Financial Studies*, 33(4), 1673–1736.
- De Bondt, W.F.M. & Thaler, R. (1985). Does the stock market overreact? *Journal of Finance*, 40(3), 793–805.
- Fama, E.F. & French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fama, E.F. & French, K.R. (2015). A five-factor model of expected stock returns. *Journal of Financial Economics*, 116(1), 1–22.
- Frazzini, A. & Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- Grinblatt, M. & Han, B. (2005). Prospect theory, mental accounting, and momentum. *Journal of Financial Economics*, 78(2), 311–339.
- Hong, H. & Stein, J.C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6), 2143–2184.
- Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3), 881–898.
- Jegadeesh, N. & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Lakonishok, J., Shleifer, A. & Vishny, R.W. (1994). Contrarian investment, extrapolation, and risk. *Journal of Finance*, 49(5), 1541–1578.
- Mitchell, M., Pedersen, L.H. & Pulvino, T. (2007). Slow moving capital. *American Economic Review*, 97(2), 215–220.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1), 1–28.
- Titman, S., Wei, K.C.J. & Xie, F. (2004). Capital investments and stock returns. *Journal of Financial and Quantitative Analysis*, 39(4), 677–700.